

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10 – SUB-STRAND 3.1 ANIMAL NUTRITION: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving Question: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive? Use the Claim–Evidence–Reasoning (CER) approach in each section. Write your answers in the space under each prompt. Use evidence from the lessons: the Kakamega maize farm (grasshopper, aphid), the mosquito, Lake Nakuru (flamingo, fish eagle), the Maasai Mara (lion, cow), your ugali and sukuma wiki model, and your food tests on chewed ugali. Name specific structures and say what each one does. Aim for clear, complete sentences. Do not just list facts: explain how and why.

Part 1: Structure, Function and Food in the Anchoring Phenomenon

At the Kakamega maize farm, grasshoppers leave ragged holes in maize leaves, aphids make the plant wilt without leaving holes, and a mosquito draws your blood. (a) Write a claim that explains why a mosquito can drink blood but a grasshopper cannot. (b) Give at least two pieces of structural evidence from the mouthparts of these insects. (c) Use reasoning to link the structure of each mouthpart (including the aphid's) to its function and to the food it can reach. Use the words adaptation and mechanical digestion where they fit.

Claim: A mosquito can drink blood but a grasshopper cannot because each insect's mouthparts are structurally adapted to a different kind of food. The mosquito's mouthparts can pierce skin and suck liquid, while the grasshopper's can only cut and grind solid food.

Evidence: (1) The mosquito has a needle-like proboscis: a multi-channel bundle of stylets (a fascicle) enclosed in a protective labium sheath. It has a salivary canal and a food canal. In our video the stylet bundle penetrated skin without tearing it. (2) The grasshopper has hard, toothed mandibles that move sideways to cut and grind. This is why it leaves ragged, torn holes in maize leaves. (3) The aphid also has a stylet bundle in a labium sheath. It pierces maize stems to reach phloem sap and wilts the plant without leaving a hole.

Reasoning: Structure decides function, and function decides which food an animal can reach. The mosquito's stylet bundle slides into tissue without tearing it, so it reaches liquid blood in capillaries. Saliva from the salivary canal helps stop the blood clotting, and the blood is drawn up the food canal. The aphid's stylet bundle works the same way on a plant: it reaches phloem sap, which is drawn up the food canal, so the plant wilts but no holes appear. The grasshopper's mandibles are hard and serrated, which suits mechanical digestion of solid leaf tissue. They have no channel for sucking liquid and they cannot form a fine needle, so a grasshopper cannot take blood. The mosquito and aphid cannot chew solid leaf. These structures are adaptations: features that help the organism survive by getting the nutrients it needs.

This answers the anchoring question. The shape of a mouthpart matches the physical form of the food (liquid or solid).

Part 2: Beaks, Teeth and the Pattern Across Animals

At Lake Nakuru, flamingos sweep their bent beaks through the water while a fish eagle tears apart a fish. At the Maasai Mara, a lion feeds on a wildebeest and a cow chews grass. (a) Choose TWO animals from this list and make a claim about how each one's feeding structure is matched to its food. (b) Give structural evidence for each (name specific features). (c) Explain why each animal could not simply swap structures with the other, and state the pattern that links these animals to the insects in Part 1.

Claim: The flamingo's beak is adapted for filtering tiny food from water, and the fish eagle's beak is adapted for gripping and tearing flesh. (A second pairing is also acceptable: the lion's teeth are adapted for gripping and shearing meat, while the cow's teeth are adapted for grinding grass.)

Evidence: The flamingo's bent beak is held upside down and lined with comb-like lamellae. Its tongue pumps water through the beak, and the lamellae act as a sieve that traps algae and small crustaceans from Lake Nakuru's water. The fish eagle has a sharply hooked, pointed beak that grips and tears fish flesh. For the mammals: the lion has long canines and carnassial teeth for gripping and slicing muscle. The cow has a dental pad and broad, ridged molars for grinding tough grass.

Reasoning: The flamingo's lamellae are good at separating very small particles from water, but they could not grip or tear a fish. The fish eagle's hooked beak can tear flesh but has no sieve, so it could not collect algae from the water. A lion's shearing teeth would not grind grass well, and a cow's flat molars could not slice meat. Each structure is shaped to match the food source and habitat, so swapping them would leave the animal unable to get its nutrients.

Pattern: Across the grasshopper's mandibles, the mosquito's and aphid's stylets, the flamingo's lamellae, the fish eagle's hooked beak and the teeth of the lion and cow, structure determines function, and function determines which food (and so which nutrients) the animal can reach.

Part 3: Inside the Body – Digesting Ugali and Sukuma Wiki

In our lessons we followed a meal of ugali and sukuma wiki through the human digestive system. We chewed a small piece of plain ugali for two full minutes without swallowing, then tested a sample of the ugali before chewing and after chewing with iodine and with Benedict's reagent (heated). Our results are shown below.

Sample	Iodine test	Benedict's test (after heating)
Ugali sample	Blue-black	Blue

Claim: During the two minutes of chewing, an enzyme in saliva acted on the starch in the ugali and broke it down into reducing sugar.

Evidence: Before chewing, the ugali turned blue-black with iodine, which shows starch is present, and stayed blue with Benedict's reagent, which shows no reducing sugar. After chewing, the ugali turned yellow-brown with iodine, so starch was no longer detected, and it turned brick-red with Benedict's reagent after heating, so reducing sugar was now present. The ugali also tasted sweet after two minutes of chewing.

Reasoning: Saliva contains amylase, a biological catalyst (enzyme) that breaks starch, a large insoluble carbohydrate, into small soluble reducing sugars. Chewing mixes the ugali with saliva and gives the amylase the two minutes it needed to act. Iodine could no longer find starch, and Benedict's reagent found sugar that was not there before. Enzymes are specific to their substrate and are not used up. Digestion continues down the digestive system: the stomach has muscular walls that churn the food (mechanical digestion), and pepsin in the stomach breaks down protein. In the small intestine, amylase in pancreatic juice continues the chemical breakdown of starch into sugars, and lipase breaks fats down into fatty acids and glycerol. These steps make the food molecules small and soluble enough to be absorbed.

1		
Ugali sample 2	Yellow-brown	Brick-red

(a) Sample 1 was tested before chewing and sample 2 after two minutes of chewing. Make a claim about what happened to the starch during the two minutes of chewing. (b) Use the table as evidence. (c) Explain the reasoning, naming the enzyme and the substance in the mouth that contains it, and describe the roles of the stomach and small intestine in digesting the meal. Which enzymes act on protein and on fat?

Part 4: Absorption, Assimilation and Health

(a) Make a claim about how the structure of the small intestine (ileum) is adapted for absorbing nutrients. (b) Give evidence from the villus diagram, including which nutrients enter the blood capillaries and which enter the lacteal. (c) Explain the difference between absorption and assimilation. Then use the Kenyan health examples (kwashiorkor in Kilifi County, iron-deficiency anaemia in adolescent girls in Nairobi) to explain why healthy villi are not enough on their own to prevent malnutrition.

Claim: The ileum is structurally adapted for efficient absorption because its villi and microvilli create a very large surface area, with thin walls and a rich blood supply.

Evidence: Villi and microvilli are finger-like projections that greatly increase the surface area. The villus wall is thin (epithelial cells), so molecules cross a short distance. Each villus has blood capillaries that receive glucose and amino acids, which travel to the liver in the hepatic portal vein. Each villus also has a central lacteal that receives fatty acids and glycerol, which recombine into fat droplets and travel in the lymphatic system.

Reasoning: A large surface area, thin walls and a good blood supply make absorption fast and efficient. Absorption is the movement of digested nutrients into the blood or lymph. Assimilation is when body cells actually use those nutrients, for example glucose in respiration and amino acids to build proteins. Malnutrition shows that structure alone is not enough: even healthy villi cannot absorb nutrients that are not in the diet. Children in Kilifi County can develop kwashiorkor from a protein-deficient diet, and many adolescent girls in Nairobi have iron-deficiency anaemia. In both cases body cells are deprived of nutrients they need, so a good diet is needed as well as a well-adapted digestive system.

Part 5: Final Explanation of the Driving Question

Now answer the driving question in one connected explanation: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive? Your answer must (a) state a clear claim, (b) use at least four pieces of evidence from different lessons (include at least one external feeding structure and at least one internal structure), (c) give reasoning that links structure to function to food to nutrients, and (d) explain how a mosquito's feeding differs from a human's in terms of what has to happen after the food is taken in. Finish by choosing one of githeri, sukuma wiki or mandazi and saying what nutrient benefit you would weigh against its cost when choosing food.

Claim: Every animal's feeding structures are adapted to the food it eats, so structure enables nutrient acquisition. The shape of a structure determines its function, and its function determines which nutrients the animal can reach.

Evidence: (1) The aphid's needle-like stylet pierces phloem in maize stems in Kakamega and draws up sap without making holes. (2) The grasshopper's hard, toothed mandibles cut and grind solid maize leaf, leaving ragged holes. (3) The flamingo's comb-like lamellae filter algae and small crustaceans from Lake Nakuru's water, while the fish eagle's hooked beak tears fish. (4) The lion's canines and carnassials and the cow's broad molars show the same match between teeth and diet. (5) Inside the body, amylase turned starch in chewed ugali into sugar (iodine yellow-brown, Benedict's brick-red), and the villi of the ileum absorb glucose and amino acids into the blood and fatty acids and glycerol into the lacteal.

Reasoning: External structures decide what food an animal can take in: needle-like stylets reach liquid, mandibles process solid tissue, lamellae sieve tiny particles, and hooked beaks tear flesh. A grasshopper cannot pierce skin and a mosquito cannot chew leaves, because each structure only works for the food it is built for. After intake, the food still has to be digested and absorbed. The mosquito feeds on liquid blood, which is already dissolved and nutrient-rich, so it needs far less processing: no chewing or grinding is needed, though the blood proteins must still be chemically digested into amino acids before they are absorbed. A human eating solid, mixed food such as ugali and sukuma wiki needs much more processing: teeth for mechanical digestion, enzymes (amylase, pepsin, lipase), the muscular stomach and the long small intestine with villi to break the food into small soluble molecules and absorb them. Once absorbed, the nutrients are assimilated by body cells for energy, growth and repair. From beak tip to intestinal microvillus, structure serves the same purpose: obtaining nutrients for survival.

Food choice: Githeri (maize and beans) supplies starch for energy and protein from the beans, and it is usually cheap. Sukuma wiki supplies vitamins and iron at low cost. Mandazi mostly supplies starch and fat with fewer nutrients. Choosing foods with more nutrients for the cost, such as githeri with sukuma wiki, helps the body get what it needs and helps prevent malnutrition.

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of structure–function–food reasoning	Gives a clear, accurate claim in every section and consistently links a named structure to its function and to the food or nutrient it allows access to, explaining why other structures could not do the same job.	Gives mostly clear claims and links structure to function, but the link to food or nutrients is sometimes missing or incomplete.	Claims are vague or missing; structures are named without explaining what they do or why they suit the food.
Use of evidence from the lessons	Uses specific, accurate evidence from several lessons (Kakamega insects, Lake Nakuru birds, Maasai Mara mammals,	Uses accurate evidence from some lessons but may rely on one or two examples or give evidence without detail.	Evidence is general, inaccurate or not drawn from the lessons.

	food-test results, villus diagram) including at least one external and one internal structure.		
Accuracy of digestion and absorption science	Correctly explains amylase action with food-test results, the roles of pepsin and lipase, villus adaptations, the two absorption routes, and absorption versus assimilation.	Mostly accurate but with minor gaps, such as a missing enzyme role or one absorption route.	Contains major errors or confuses digestion, absorption and assimilation.
Connecting external feeding structures to internal digestion	Explains clearly why a mosquito's liquid food and a human's mixed diet need different amounts of processing, and ties all parts into one unifying explanation.	Makes a connection between external and internal structures but does not fully explain the difference in processing.	Treats external and internal structures as unrelated topics.
Scientific communication and Kenyan context	Writes in organised, precise sentences with correct terms (adaptation, mechanical digestion, enzyme, villi, assimilation) and applies the biology sensibly to Kenyan food and health examples.	Writing is mostly clear with some correct terms; Kenyan examples are used but with limited explanation.	Writing is hard to follow, terms are misused, and Kenyan examples are absent or unconnected to the biology.

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